NATIONAL INSTITUTE OF TECHNOLOGY PATNA
END-SEMESTER EXAMINATION, MARCH 2025

Program: B.Tech. (CSE) **Semester:** 2nd
Department: CSE **Course Code:** CS24107
Full Marks: 60 **Course Name:** Web Technology
Duration of Examination: 3 hours

INSTRUCTIONS
Answer all the questions. Assume Missing data, if any.

1. [1x6=6M] Write the CSS code for the following: [CO2][L3]
   a. Using CSS animation specify that the animation of the element should alternate between running forwards and backwards.
   b. Insert an image before, and after <p> elements
   c. Using the transition shorthand property, specify width changes for the element should have: "2" second duration, "ease-in-out" speed curve, and a "0.5" second delay before starting.
   d. The animation of the element should have a "ease-in-out" speed curve.
   e. The animation of the element should alternate between running forwards and backwards.
   f. Using CSS change both the background-color and the position of the element when the animation is 25% complete, 50% complete, and again when the animation is 100% complete

2. [1x10=10M] Write jQuery code for the following:[CO2][CO3][CO1][L2][L3]
   a. If you press a keyboard key inside an <input> element, it should be hidden. Use the correct event to do so.
   b. Use the correct selector to hide all odd table rows in a table.
   c. Use a jQuery method to toggle between adding and removing the "important" class from the <p> element on click.
   d. Use proper jQuery method to set the following styles for <p>: Green background color, white text color, 25 pixels font size and a padding of 15 pixels.
   e. Use a jQuery method to remove all elements with class="test" and class="demo".
   f. Use a jQuery method to insert the text "YES!" at the beginning of a <p> element.
   g. Use a jQuery method to toggle between fading in and out a element, when clicking on a button. The duration of the effect should be 1000 milliseconds.
   h. Use a jQuery method to toggle between sliding up and down a element, when clicking on a button.
   i. Create a HTML input field which should take Name (combination of First and Last name) as an input , the length of name should not be more than 20 characters also the input field should show a message "enter your name".
   j. Write a JavaScript function that checks if a cookie is set. If the cookie is set it will display a greeting. If the cookie is not set, it will display a prompt box, asking for the name of the user, and stores the username cookie for 365 days

3. [3M] Write an AJAX script to send HTML form data (say First name and Last name) to the server.[CO3][L3]

4. [7M] Imagine there are two valid users, U1 and U2, registered to a website that provides profile creation, viewing, and updating services. Users are able to update their profiles only when they are logged in to the website. ideally each user should be able to update their own profile only. However a methods ~~descover~~ discover

bug due to which U1 was able to update the profile of U2 and vice versa. Explain with an example what would have gone wrong and suggest a fix for the same.[CO1][L4]

5. [5M] Write a PHP script to create, retrieve and modify a Cookie Value. [CO4] [L3]

6. [10M]

a. Write a PHP script that reads name, rollnumber and email from a HTML Form. [CO4] [L3]

b. Write a PHP script to upload a image file of size less than equal to 20 MB and acceptable image format is either jpeg/jpg/gif/png. [CO4] [L3]

OR(For both 6.a and 6.b)

Write a PHP script that reads the name, roll number, and email from an HTML form. Upon submission, the script should check if any of the fields are empty. If so, it should reload the form with a message next to each empty field stating "This field cannot be empty," while the non-empty fields should retain the submitted values. If all fields are filled, the user should be redirected to a page called response.html, where the submitted values are displayed. [CO4] [L3]

7. [1*10 = 10M] Predict the output of the following code. Please provide proper reasoning, without which you may not get marks:[CO2][CO3][CO4][L2]

a. div {border-width:5px; border-style: dotted solid double dashed;}	b. span {border: 1px solid red; outline: green dotted thick; }

c. let x = "5";let y = 2;
console.log(x + y);
console.log(x - y);
	d. let x = "hello"; let y = new String("hello");
	console.log(x = = y);
	console.log(x = = = y);

e. let x = "b"; let y = "a"; console.log(x + y + + y + y);	f. <?php $input = "Swapna<td>Lawrence</td>you are reallypretty!"; $input = strip_tags($input,""); echo $input; ?>
g. <?php $a = "$winner"; $b = "$looser"; echo $a, $b; ?>	h. <?php function fun() {static $x = 0; echo $x; $x++;} fun(); fun(); fun(); ?>
i. <?php $x = 3, 4, 5, 6; echo "$x"; ?>	j. <?php $x; if ($x) print "hi"; else print "how are u"; ?>

8. [9M]An ISP is granted a block of addresses starting with 172.27.16.0/16. The ISP wants to distribute these blocks to 2600 customers as follows: [CO1][L3]

a. The first group has 200 medium-size businesses; each needs approximately 128 addresses

b. The second group has 400 small businesses; each needs approximately 16 addresses.

c. The third group has 2000 households; each needs 4 addresses